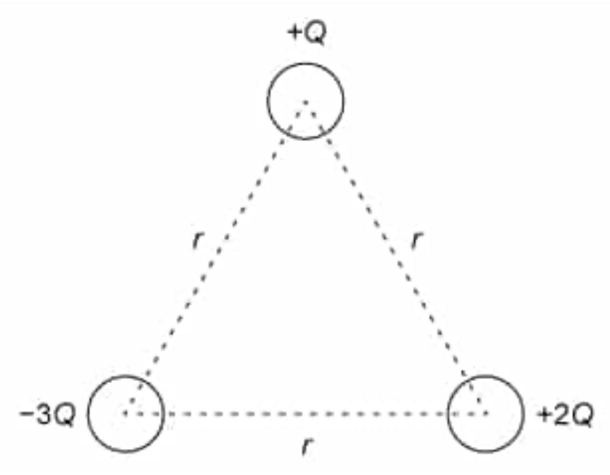
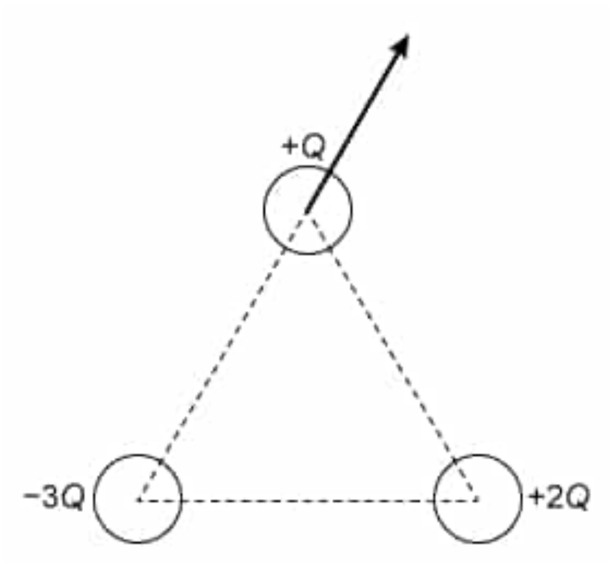
Three charged spheres are each held fixed at a distance *r* from the other two as shown in the following figure.



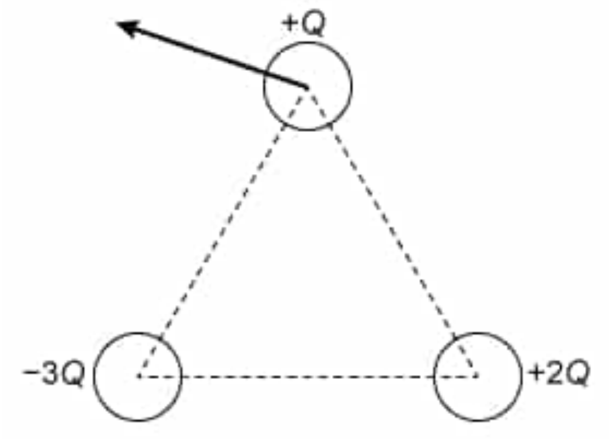
The diameter of each sphere is much smaller than *r*. The sphere of charge +Q is then allowed to move freely. Which of the following figures best illustrates the acceleration of the sphere with charge +Q immediately after it is released?

A.



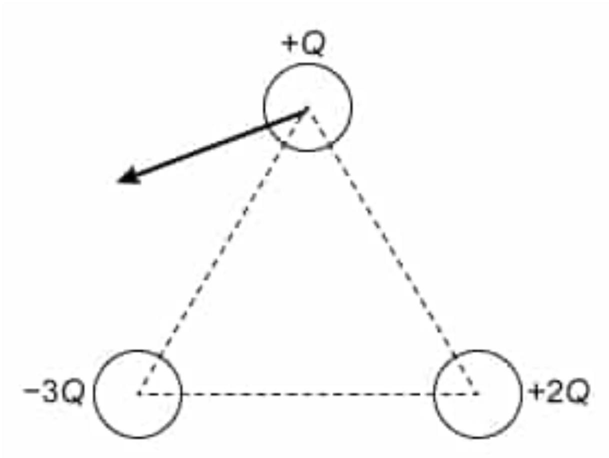
[2%]

B.



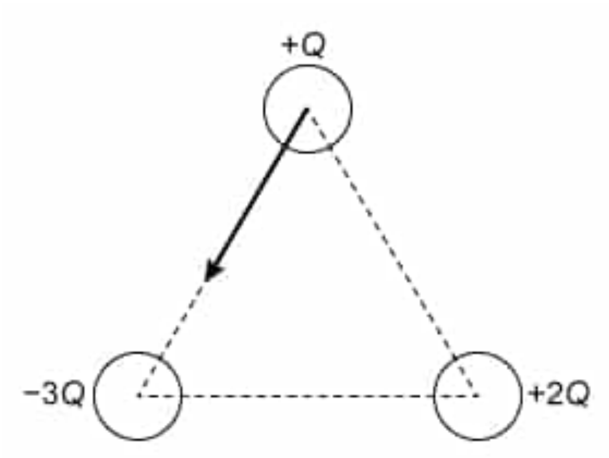
[4%]

C.



[77%]

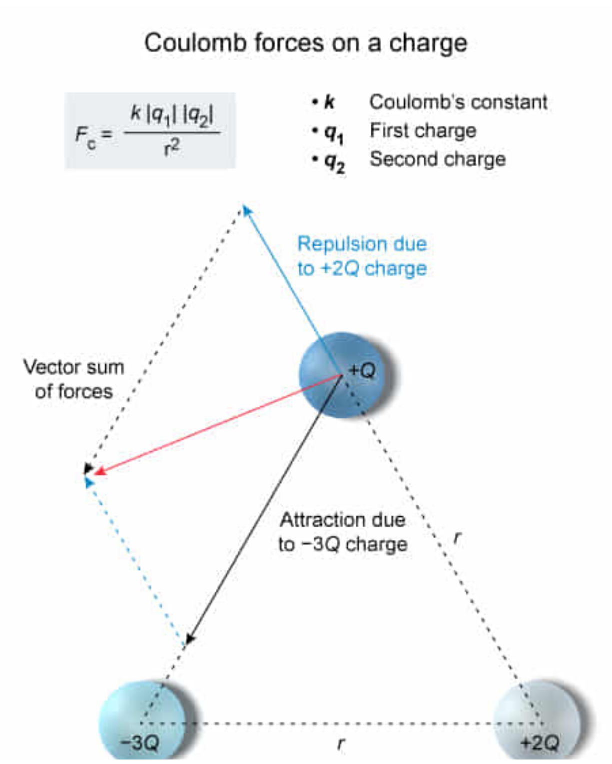
○ D.



[14%]

Correct
77% answered correctly

Explanation:



Stationary charged objects interact via **electrostatic forces**. Charges of the same sign repel and charges of opposite sign attract. According to Coulomb's law, the magnitude of the electrostatic force between two charges (q_1 and q_2) is:

$$F_c = \frac{k|q_1||q_2|}{r^2}$$

where r is the distance between the charges and k is Coulomb's constant, $k = 9.0 \times 10^9\ \text{N·m}^2/\text{C}$.

In this question, the sphere of charge +Q experiences two unique electrostatic forces because of interactions with the other charged spheres. First, the sphere of charge +2Q and the sphere of charge +Q exert on each other a repulsive force that acts along the line between each, and the magnitude of this force is:

$$F_{+2Q} = \frac{k(Q)(2Q)}{r^2}$$

In addition, the sphere of charge −3Q and sphere of charge +Q exert on each other an attractive force of even greater magnitude:

$$F_{-3Q} = \frac{k(Q)(3Q)}{r^2}$$

When the two force vectors are added together, the result points downward and to the left from +Q's original position. Since Newton's second law implies acceleration a is proportional to the net force on an object:

$$\vec{a} = \frac{\sum \vec{F}}{m}$$

the best representation of the acceleration of the +Q charge after release is the vector shown in Choice C.

(Choice A) This answer incorrectly assumes the force between +Q and −3Q is repulsive. Instead, the electrostatic force between these spheres is attractive.

(Choice B) This answer incorrectly assumes the magnitude of the force between +Q and +2Q is greater than that between +Q and −3Q.

(Choice D) This answer incorrectly assumes the force between +Q and +2Q is attractive. Instead, the electrostatic force between these spheres is repulsive.

Educational objective:
The direction and magnitude of the electrostatic force between charges depends on the relative signs of the charges, the magnitude of the charges, and the distance

between them. If multiple forces act on an object, the net force is determined by vector addition.

Time spent: 0 sec	**QID**:403925
Subject: Physics	**System:** Electrostatics and Circuits